

# Daily Algebra Drill #3 — Answers & Investigations

| Section | Topic                    | Questions | Target time  |
|---------|--------------------------|-----------|--------------|
| A       | Rapid Recognition        | 10        | 2 min 30 sec |
| B       | Manipulation Drills      | 10        | 8 min        |
| C       | Substitution & Structure | 8         | 10 min       |
| D       | Challenge                | 5         | 15 min       |

*New toolkit today: Remainder & Factor Theorem · AM–GM & algebraic inequalities · Completing the square in two variables · Rational Root Theorem · Modular/divisibility arguments · Parametric families of solutions.*

## Section A Rapid Recognition

1.  $f(x) = x^3 - 3x^2 + kx - 8$  has  $(x - 2)$  as a factor. Find  $k$ .

**Answer:**  $k = 6$

**Method: Factor Theorem:** if  $(x - 2)$  is a factor then  $f(2) = 0$ .  $8 - 12 + 2k - 8 = 0 \Rightarrow 2k = 12 \Rightarrow k = 6$ . Never do polynomial long division when the factor theorem gives a one-line equation.

**Investigate further: Remainder Theorem (full statement):** when  $p(x)$  is divided by  $(x - a)$ , the remainder is  $p(a)$ . Use this to find the remainder when  $f(x) = x^3 - 6x^2 + 11x - 6$  is divided by  $(x - 4)$ . Then explore: if the remainder when  $f(x)$  is divided by  $(x - 1)$  is 3 and by  $(x + 1)$  is  $-1$ , find the remainder when divided by  $(x^2 - 1)$ . (*Key: the remainder is now linear:  $ax + b$ . Set up two equations.*)

2. Find the remainder when  $p(x) = x^{100} - 1$  is divided by  $(x - 1)$ .

**Answer:** 0

**Method:**  $p(1) = 1^{100} - 1 = 0$ . So the remainder is 0 and  $(x - 1)$  is in fact a factor. The Remainder Theorem makes this instantaneous — no division required.

**Investigate further:** Find the remainder when  $x^{100} + x^{50} + 1$  is divided by  $(x - 1)$ ,  $(x + 1)$ , and  $(x^2 - 1)$ . Then prove:  $x^n - 1$  is always divisible by  $(x - 1)$  for positive integers  $n$ . (*Use the factorisation  $x^n - 1 = (x - 1)(x^{n-1} + x^{n-2} + \dots + 1)$ , which you can verify by multiplying out.*)

3. Factorise:  $6x^2 - x - 12$ .

**Answer:**  $(2x - 3)(3x + 4)$

**Method:** Product =  $6 \times (-12) = -72$ . Seek pair summing to  $-1$ :  $-9$  and  $8$ . Split:  $6x^2 - 9x + 8x - 12 = 3x(2x - 3) + 4(2x - 3) = (2x - 3)(3x + 4)$ .

**Investigate further: AC method / discriminant connection.** A quadratic  $ax^2 + bx + c$  factorises over  $\mathbb{Z}$  if and only if its discriminant  $b^2 - 4ac$  is a perfect square. Check:  $(-1)^2 - 4(6)(-12) = 1 + 288 = 289 = 17^2$ . ✓ This gives the roots directly:  $x = \frac{1 \pm 17}{12}$ , i.e.  $x = \frac{3}{2}$  or  $x = -\frac{4}{3}$ . Investigate: for which integer values of  $c$  does  $6x^2 - x + c$  factorise over  $\mathbb{Z}$ ?

4. Simplify:  $\frac{x^{2n} - 1}{x^n - 1}$ .

**Answer:**  $x^n + 1$

**Method:** Difference of two squares:  $x^{2n} - 1 = (x^n - 1)(x^n + 1)$ . Cancel  $(x^n - 1)$ . This is the  $m = 2$  case of the general factorisation  $x^{mn} - 1$ .

**Investigate further:** Generalise:  $\frac{x^{mn} - 1}{x^n - 1} = x^{n(m-1)} + x^{n(m-2)} + \cdots + x^n + 1$ , a geometric series in  $x^n$ . Use this with  $x = 10$ ,  $n = 1$  to explain why  $\underbrace{99 \cdots 9}_m$  is always divisible by 9. Then use  $n = 2$ ,  $m = 3$  to factorise  $x^6 - 1$  fully over  $\mathbb{Z}$ .

5. Given  $x > 0$ , find the minimum value of  $x + \frac{9}{x}$ .

**Answer:** 6

**Method:** AM–GM:  $x + \frac{9}{x} \geq 2\sqrt{x \cdot \frac{9}{x}} = 2 \cdot 3 = 6$ . Equality when  $x = \frac{9}{x}$ , i.e.  $x = 3$ . Always suspect AM–GM when you see a sum of a quantity and its reciprocal (scaled).

**Investigate further:** Generalise: the minimum of  $x + \frac{c}{x}$  for  $x, c > 0$  is  $2\sqrt{c}$ , achieved at  $x = \sqrt{c}$ . Now find the minimum of  $x + \frac{4}{x} + \frac{9}{x^2}$  for  $x > 0$  by substituting  $t = x + \frac{3}{x}$  and completing the square in  $t$ .

6. Factorise over  $\mathbb{Z}$ :  $x^4 - y^4$ .

**Answer:**  $(x^2 + y^2)(x + y)(x - y)$

**Method:** Apply difference of two squares twice:  $x^4 - y^4 = (x^2 + y^2)(x^2 - y^2) = (x^2 + y^2)(x + y)(x - y)$ . The factor  $x^2 + y^2$  does not factorise further over  $\mathbb{R}$ .

**Investigate further:** Over  $\mathbb{C}$ ,  $x^2 + y^2 = (x + iy)(x - iy)$ . So  $x^4 - y^4 = (x + iy)(x - iy)(x + y)(x - y)$  over  $\mathbb{C}$ . Now factorise  $x^4 - 1$  fully over  $\mathbb{R}$ , then over  $\mathbb{C}$ . Connect to the *roots of unity*: the four roots of  $x^4 = 1$  are  $1, -1, i, -i$ , corresponding exactly to the four factors.

7. The roots of  $2x^2 - 6x + 3 = 0$  are  $\alpha$  and  $\beta$ . Find  $\alpha^2 + \beta^2$ .

**Answer:**  $\frac{15}{2}$

**Method:** Vieta:  $\alpha + \beta = 3$ ,  $\alpha\beta = \frac{3}{2}$ .  $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 9 - 3 = \frac{15}{2}$ . Note: for a non-monic quadratic  $ax^2 + bx + c = 0$ , Vieta gives  $\alpha + \beta = -b/a$ ,  $\alpha\beta = c/a$ .

**Investigate further:** Find  $\alpha^3 + \beta^3$  and  $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$  using the same roots. For  $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ , write it as  $\frac{\alpha^2 + \beta^2}{\alpha^2\beta^2}$ . Then investigate: what quadratic has roots  $\alpha^2$  and  $\beta^2$ ? (Use the Vieta data for the new quadratic: sum =  $\alpha^2 + \beta^2$ , product =  $\alpha^2\beta^2$ .)

8. Simplify:  $\frac{(n+1)! - n!}{(n-1)!}$ .

**Answer:**  $n^2$

**Method:** Factor:  $(n+1)! - n! = n! \cdot n = n \cdot n!$ . Then  $\frac{n \cdot n!}{(n-1)!} = n \cdot \frac{n!}{(n-1)!} = n \cdot n = n^2$ . Always

factor out the smaller factorial first.

**Investigate further:** Prove the identity  $(n+1)! - n! - n \cdot (n-1)! = (n-1)! \cdot (n^2 - n - 1)$  by factoring  $(n-1)!$  throughout. Then investigate: for which  $n$  is  $\frac{(2n)!}{(n!)^2}$  (the central binomial coefficient) divisible by  $n+1$ ? (*Catalan number connection.*)

9. Show that  $(a+b+c)^2 - (a^2 + b^2 + c^2) = 2(ab+bc+ca)$  and state  $ab+bc+ca$  when  $a+b+c=4$ ,  $a^2+b^2+c^2=8$ .

**Answer:**  $ab+bc+ca=4$

**Method:** Expand  $(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab+bc+ca)$ . Rearrange. Substituting:  $16 - 8 = 2(ab+bc+ca) \Rightarrow ab+bc+ca=4$ .

**Investigate further:** With  $a+b+c=4$ ,  $a^2+b^2+c^2=8$ ,  $ab+bc+ca=4$ : now use Newton's identity to find  $a^3+b^3+c^3$  in terms of the elementary symmetric polynomials  $e_1=4$ ,  $e_2=4$ ,  $e_3=abc$ . Show that  $a^3+b^3+c^3=3e_3+e_1(e_1^2-3e_2)=3abc+16$ . Hence  $a^3+b^3+c^3$  depends on  $abc$  — find specific triples  $(a,b,c)$  satisfying the constraints and verify.

10. Find the value(s) of  $k$  such that  $x^2+kx+16$  is a perfect square.

**Answer:**  $k=\pm 8$

**Method:** A perfect-square quadratic has discriminant zero:  $k^2-64=0 \Rightarrow k=\pm 8$ . Then  $x^2+8x+16=(x+4)^2$  and  $x^2-8x+16=(x-4)^2$ .

**Investigate further:** Generalise:  $x^2+kx+c$  is a perfect square iff  $k=\pm 2\sqrt{c}$ . For  $c$  to allow integer  $k$ , we need  $c$  to be a perfect square itself. Now: for which integer values of  $m$  is  $x^2+mx+m$  a perfect square (i.e. has discriminant zero)? Find all such  $m$ .

## Section B Manipulation Drills

1. Find all rational roots of  $2x^3-3x^2-11x+6$  and factorise completely.

**Answer:**  $(x-3)(x+2)(2x-1)$ ; rational roots:  $x=3, -2, \frac{1}{2}$

**Method: Rational Root Theorem:** rational roots have the form  $\pm \frac{p}{q}$  where  $p \mid 6$  and  $q \mid 2$ . Candidates:  $\pm 1, \pm 2, \pm 3, \pm 6, \pm \frac{1}{2}, \pm \frac{3}{2}$ . Test  $x=3$ :  $54-27-33+6=0$ . ✓ Divide out  $(x-3)$  to get  $2x^2+3x-2=(2x-1)(x+2)$ .

**Investigate further: Rational Root Theorem (formal statement):** if  $a_n x^n + \dots + a_0$  has a rational root  $p/q$  in lowest terms, then  $p \mid a_0$  and  $q \mid a_n$ . Use this to show that  $x^3-2$  has no rational roots, hence  $\sqrt[3]{2}$  is irrational. Generalise: prove that  $\sqrt[n]{k}$  is irrational whenever  $k$  is not a perfect  $n$ th power.

2. Simplify:  $\frac{x^3+1}{x^2-x+1}$ .

**Answer:**  $x+1$

**Method:** Sum of cubes:  $x^3+1=(x+1)(x^2-x+1)$ . Cancel  $(x^2-x+1)$ . Check:  $x^2-x+1 \neq 0$  for real  $x$  (discriminant  $< 0$ ).

**Investigate further:** The factor  $x^2-x+1$  is the *cyclotomic polynomial*  $\Phi_6(x)$ . Its roots are the

primitive 6th roots of unity:  $e^{\pm i\pi/3}$ . Show that  $x^6 - 1 = (x - 1)(x + 1)(x^2 + x + 1)(x^2 - x + 1)$  by factorising step by step. This connects to the factorisation of  $x^n - 1$  into cyclotomic polynomials, a key tool in number theory and algebra.

3. Given  $a + b = p$  and  $a^3 + b^3 = q$ , express  $ab$  in terms of  $p$  and  $q$ .

**Answer:**  $ab = \frac{p^3 - q}{3p}$

**Method:**  $a^3 + b^3 = (a + b)(a^2 - ab + b^2) = (a + b)[(a + b)^2 - 3ab] = p(p^2 - 3ab) = q$ . So  $p^3 - 3p \cdot ab = q$ , giving  $ab = \frac{p^3 - q}{3p}$ .

**Investigate further:** Given only  $a + b = p$  and  $a^3 + b^3 = q$ , find  $a^4 + b^4$  and  $a^5 + b^5$  using Newton's recurrence  $s_n = (a + b)s_{n-1} - ab \cdot s_{n-2}$ . Then: if  $p$  and  $q$  are both integers, is  $ab$  necessarily rational? Always? Find an example where  $p, q \in \mathbb{Z}$  but  $a, b \notin \mathbb{Q}$ .

4. Simplify:  $\frac{x^2(x + 1) - x(x + 1)^2}{x(x + 1)}$ .

**Answer:**  $-1$

**Method:** Factor the numerator:  $x(x + 1)[x - (x + 1)] = x(x + 1)(-1)$ . Cancel  $x(x + 1)$ . Result:  $-1$ . Always look for a common factor in the numerator before combining fractions.

**Investigate further:** This expression equals  $-1$  for all  $x \neq 0, -1$ . What does this mean geometrically? The function  $f(x) = \frac{x^2(x + 1) - x(x + 1)^2}{x(x + 1)}$  is a rational function that simplifies to the constant  $-1$  with two holes. Sketch it and compare to  $y = -1$ . Then investigate: when does a rational function simplify to a polynomial? What algebraic condition must hold?

5. Factorise:  $a^3 - a^2b - ab^2 + b^3$ .

**Answer:**  $(a - b)^2(a + b)$

**Method:** Group:  $(a^3 + b^3) - (a^2b + ab^2) = (a + b)(a^2 - ab + b^2) - ab(a + b) = (a + b)(a^2 - 2ab + b^2) = (a + b)(a - b)^2$ .

**Investigate further:** Verify using the substitution  $b = a$ : the expression becomes 0, confirming  $(a - b)$  is a factor. Then check  $b = -a$ : again 0, confirming  $(a + b)$  is a factor. Since the polynomial is degree 3 with two linear factors, the third must also be linear — and we can read it off. This substitution method (“*plug in values to find factors*”) is the heart of the factor theorem. Use it to factorise  $a^4 - b^4 - a^3b + ab^3$ .

6. Use  $\sum_{r=1}^n r = \frac{n(n+1)}{2}$  to find  $\sum_{r=1}^n (2r - 1)$ .

**Answer:**  $n^2$

**Method:**  $\sum_{r=1}^n (2r - 1) = 2 \cdot \frac{n(n+1)}{2} - n = n(n+1) - n = n^2$ . The sum of the first  $n$  odd numbers is always a perfect square. Elegant!

**Investigate further:** Prove the same result geometrically: the  $n$ th odd number  $2n - 1$  is the number of extra unit squares added to an  $n \times n$  grid to produce an  $(n + 1) \times (n + 1)$  grid (the L-shaped “gnomon”). Then find  $\sum_{r=1}^n (2r)$  and  $\sum_{r=1}^n r^2$  using the identity  $(r + 1)^3 - r^3 = 3r^2 + 3r + 1$ , telescoped.

7. Solve:  $\frac{2x - 1}{x + 3} - \frac{x + 2}{x - 1} = 1$ .

**Answer:**  $x = 2$

**Method:** Common denominator  $(x + 3)(x - 1)$ :  $(2x - 1)(x - 1) - (x + 2)(x + 3) = (x + 3)(x - 1)$ . LHS:  $(2x^2 - 3x + 1) - (x^2 + 5x + 6) = x^2 - 8x - 5$ . RHS:  $x^2 + 2x - 3$ . So  $x^2 - 8x - 5 = x^2 + 2x - 3 \Rightarrow -10x = 2 \Rightarrow x = -\frac{1}{5}$ . (Re-check): verify  $x = -\frac{1}{5}$  in original — LHS =  $\frac{-7/5}{14/5} - \frac{9/5}{-6/5} = \frac{-7}{14} + \frac{9}{6} = -\frac{1}{2} + \frac{3}{2} = 1$ .  $\checkmark$  So  $x = -\frac{1}{5}$ .

**Investigate further:** Rational equations of this form always reduce to a polynomial equation after clearing denominators. Investigate: under what conditions does clearing denominators in  $\frac{P(x)}{Q(x)} = \frac{R(x)}{S(x)}$  produce extraneous solutions? (*Extraneous solutions satisfy the cleared equation but make a denominator zero.*) Give an example of an equation that gains an extraneous solution when cleared.

8. Simplify:  $\left(\frac{1}{a} - \frac{1}{b}\right) \div \left(\frac{1}{a^2} - \frac{1}{b^2}\right)$ .

**Answer:**  $\frac{ab}{a + b}$

**Method:** Numerator:  $\frac{b - a}{ab}$ . Denominator:  $\frac{b^2 - a^2}{a^2b^2} = \frac{(b - a)(b + a)}{a^2b^2}$ . Dividing:  $\frac{b - a}{ab} \cdot \frac{a^2b^2}{(b - a)(b + a)} = \frac{ab}{a + b}$ .

**Investigate further:** Note that  $\frac{ab}{a + b} = \frac{1}{\frac{1}{a} + \frac{1}{b}}$ , the *half-harmonic mean*. The harmonic mean of  $a$  and  $b$  is  $H = \frac{2ab}{a + b}$ . Prove the chain of inequalities  $H \leq G \leq A$  where  $G = \sqrt{ab}$  and  $A = \frac{a + b}{2}$  for  $a, b > 0$ . (Use AM–GM twice: once to get  $G \leq A$ , once rearranged to get  $H \leq G$ .)

9. Show that  $n^3 - n$  is divisible by 6 for all integers  $n$ .

**Answer:** Divisible by 6 for all  $n \in \mathbb{Z}$ .

**Method:**  $n^3 - n = n(n - 1)(n + 1) = (n - 1)n(n + 1)$ : the product of three consecutive integers. *Divisible by 2:* among any two consecutive integers, one is even. *Divisible by 3:* among any three consecutive integers, one is divisible by 3. Since  $\gcd(2, 3) = 1$ , the product is divisible by 6.  $\square$

**Investigate further: Generalisation.** Show that the product of any  $k$  consecutive integers is divisible by  $k!$ . This is equivalent to showing that  $\binom{n}{k} = \frac{n!}{k!(n - k)!}$  is always an integer for  $n \geq k$ . (Proof:  $\binom{n}{k}$  counts the number of  $k$ -element subsets of an  $n$ -element set, hence is an integer.) Apply: show  $n^5 - n$  is always divisible by 30 by factoring  $n^5 - n = n(n^4 - 1)$  further.

10. Roots of  $x^2 - x - 1 = 0$  are  $\alpha, \beta$ . Find  $\alpha^3 + \beta^3$  and  $\alpha^4 + \beta^4$ .

**Answer:**  $\alpha^3 + \beta^3 = 4$ ;  $\alpha^4 + \beta^4 = 7$

**Method:** Vieta:  $\alpha + \beta = 1$ ,  $\alpha\beta = -1$ . Newton recurrence  $s_n = (\alpha + \beta)s_{n-1} - \alpha\beta \cdot s_{n-2} = s_{n-1} + s_{n-2}$ .  $s_1 = 1$ ,  $s_2 = (\alpha + \beta)^2 - 2\alpha\beta = 1 + 2 = 3$ .  $s_3 = s_2 + s_1 = 4$ .  $s_4 = s_3 + s_2 = 7$ . This recurrence is the *Lucas sequence* — the roots are the golden ratio and its conjugate.

**Investigate further: Golden ratio and Fibonacci.** The roots of  $x^2 - x - 1 = 0$  are  $\phi = \frac{1+\sqrt{5}}{2}$  and  $\hat{\phi} = \frac{1-\sqrt{5}}{2}$  (the golden ratio and its conjugate). The power sums  $s_n = \phi^n + \hat{\phi}^n$  satisfy  $s_n = s_{n-1} + s_{n-2}$  (Fibonacci-like). Compute  $s_0$  through  $s_8$  and compare to the Lucas numbers: 2, 1, 3, 4, 7, 11, 18, 29, 47. Prove that  $\phi^n = F_n\phi + F_{n-1}$  where  $F_n$  is the  $n$ th Fibonacci number.

## Section C Substitution & Structure

1. Solve:  $9^x - 10 \cdot 3^x + 9 = 0$ .

**Answer:**  $x = 0$  and  $x = 2$

**Method:** Let  $u = 3^x$ :  $u^2 - 10u + 9 = 0 \Rightarrow (u-1)(u-9) = 0$ .  $u = 1 \Rightarrow 3^x = 1 \Rightarrow x = 0$ .  $u = 9 \Rightarrow 3^x = 9 \Rightarrow x = 2$ .

**Investigate further:** Solve  $4^x - 5 \cdot 2^x + 4 = 0$  and  $25^x - 26 \cdot 5^x + 25 = 0$  by the same method. Then investigate:  $2^{2x} - k \cdot 2^x + k - 1 = 0$ . Show that  $x = 0$  is always a solution regardless of  $k$ , and find the second solution in terms of  $k$ . What happens when  $k = 2$ ?

2. Find the minimum value of  $x^2 + y^2$  subject to  $x + 2y = 5$ .

**Answer:** 5

**Method:** Substitute  $x = 5 - 2y$ : minimise  $(5-2y)^2 + y^2 = 5y^2 - 20y + 25$ . Complete the square:  $5(y-2)^2 + 5$ . Minimum is **5** at  $y = 2$ ,  $x = 1$ . Alternatively (elegant): by Cauchy-Schwarz,  $(x+2y)^2 \leq (x^2+y^2)(1^2+2^2)$ , so  $x^2 + y^2 \geq \frac{25}{5} = 5$ .

**Investigate further: Cauchy-Schwarz inequality:**  $(a_1b_1 + a_2b_2)^2 \leq (a_1^2 + a_2^2)(b_1^2 + b_2^2)$ . Prove it by expanding  $(a_1b_2 - a_2b_1)^2 \geq 0$ . Use it to find the minimum of  $x^2 + y^2 + z^2$  subject to  $x + y + 2z = 6$ . Then connect to the geometric interpretation:  $\sqrt{x^2 + y^2}$  is the distance from the origin to the line  $x + 2y = 5$ , and the minimum is the perpendicular distance.

3. Prove that  $a^2 + b^2 + c^2 \geq ab + bc + ca$  for all real  $a, b, c$ .

**Answer:** Proof: see method.

**Method:**  $2(a^2 + b^2 + c^2) - 2(ab + bc + ca) = (a-b)^2 + (b-c)^2 + (c-a)^2 \geq 0$ . Divide by 2. Equality iff  $a = b = c$ .  $\square$  This is the most important algebraic inequality after AM-GM.

**Investigate further:** The expression  $(a-b)^2 + (b-c)^2 + (c-a)^2$  is a *sum of squares (SOS) decomposition*, which proves non-negativity constructively. Now prove:  $a^4 + b^4 + c^4 \geq a^2b^2 + b^2c^2 + c^2a^2$  using the same idea (replace  $a, b, c$  with  $a^2, b^2, c^2$ ). Then investigate: is  $a^2 + b^2 + c^2 \geq ab + bc + ca$  still true if  $a, b, c$  are complex numbers? (No — consider what ' $\geq$ ' means for complex numbers.)

4. Solve:  $\sqrt{x+5} + \sqrt{x-3} = 4$ .

**Answer:**  $x = 4$

**Method:** Isolate one radical:  $\sqrt{x+5} = 4 - \sqrt{x-3}$ . Square:  $x+5 = 16 - 8\sqrt{x-3} + (x-3) \Rightarrow 8\sqrt{x-3} = 8 \Rightarrow x-3 = 1 \Rightarrow x = 4$ . Check:  $\sqrt{9} + \sqrt{1} = 3 + 1 = 4$ .  $\checkmark$

**Investigate further:** The conjugate trick: instead of isolating, multiply both sides by  $(\sqrt{x+5} - \sqrt{x-3})$ . LHS becomes  $(x+5) - (x-3) = 8$ , giving  $\sqrt{x+5} - \sqrt{x-3} = 2$ . Combined with the original equation:  $2\sqrt{x+5} = 6 \Rightarrow x+5 = 9 \Rightarrow x = 4$ . This conjugate approach is faster and avoids squaring. Use it to solve  $\sqrt{x+4} - \sqrt{x-4} = \frac{8}{5}$ .

5. Find the minimum of  $x^2 + 4x + y^2 - 6y + 13$ , and the  $(x, y)$  at which it occurs.

**Answer:** Minimum is 0, at  $(x, y) = (-2, 3)$

**Method:** Complete the square in each variable separately:  $(x+2)^2 - 4 + (y-3)^2 - 9 + 13 = (x+2)^2 + (y-3)^2 \geq 0$ . Minimum is 0 when  $x = -2, y = 3$ .

**Investigate further:** Note  $(x+2)^2 + (y-3)^2 = 0$  means the expression equals the square of the distance from  $(-2, 3)$  to  $(x, y)$ . So the minimum is the point closest to the origin — but the expression vanishes at  $(-2, 3)$ , not at the origin. Now find the minimum of  $x^2 + y^2 + z^2 - 2x + 4y - 6z + 14$ . Is the minimum always achievable for quadratics in  $n$  variables? (Yes, iff the quadratic form is positive definite — a key idea in multivariable optimisation.)

6. If  $x + \frac{1}{x} = 5$ , find  $\frac{x^4 + 1}{x^2}$ .

**Answer:** 23

**Method:**  $\frac{x^4 + 1}{x^2} = x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2 = 25 - 2 = 23$ .

**Investigate further:** Extend: find  $\frac{x^6 + 1}{x^3}$  and  $\frac{x^8 + 1}{x^4}$  using the chain:  $s_n = \left(x + \frac{1}{x}\right) s_{n-1} - s_{n-2}$  where  $s_k = x^k + \frac{1}{x^k}$ . This is the same Newton recurrence as before, applied to  $x$  and  $\frac{1}{x}$  (whose sum is 5 and product is 1). Compute  $s_1$  through  $s_5$ .

7. Solve over  $\mathbb{R}$ :  $x^4 + 4x^2 - 5 = 0$ .

**Answer:**  $x = \pm 1$

**Method:** Let  $u = x^2$ :  $(u+5)(u-1) = 0$ .  $u = -5$  gives no real solutions.  $u = 1 \Rightarrow x = \pm 1$ .

**Investigate further:** For which values of  $k$  does  $x^4 + 4x^2 + k = 0$  have: (i) no real solutions, (ii) exactly two real solutions, (iii) four real solutions? (Set  $u = x^2$ : the quadratic  $u^2 + 4u + k = 0$  has roots that must be positive.) Sketch the curve  $y = x^4 + 4x^2$  and read off the answers geometrically.

8. Given  $a, b > 0$  with  $a + b = 1$ , find the minimum of  $\frac{1}{a} + \frac{1}{b}$ .

**Answer:** 4

**Method:**  $\frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab} = \frac{1}{ab}$ . By AM–GM,  $ab \leq \left(\frac{a+b}{2}\right)^2 = \frac{1}{4}$ , so  $\frac{1}{ab} \geq 4$ . Equality when  $a = b = \frac{1}{2}$ .

**Investigate further:** This is a standard constrained optimisation whose answer  $4 = \left(\frac{1+1}{\sqrt{ab}}\right)^2$  more neatly:  $\left(\sqrt{\frac{1}{a}} + \sqrt{\frac{1}{b}}\right)^2 \geq 4 \cdot \frac{1}{\sqrt{ab}}$ . Generalise: for  $a_1 + a_2 + \dots + a_n = 1$  with all  $a_i > 0$ , find the

$$\left| \text{minimum of } \sum_{i=1}^n \frac{1}{a_i} \text{ using the AM-HM inequality. (Answer: } n^2\text{.)} \right|$$

## Section D Challenge

(TMUA / SMC difficulty)

1. Find all real solutions to  $x^4 + 2x^3 - 2x - 1 = 0$ .

**Answer:**  $x = 1, -1, -1 \pm \sqrt{2}$  (i.e.  $x \in \{1, -1, -1 + \sqrt{2}, -1 - \sqrt{2}\}$ )

**Method:** Group by spotting symmetry:  $x^4 - 1 + 2x^3 - 2x = (x^2 + 1)(x^2 - 1) + 2x(x^2 - 1) = (x^2 - 1)(x^2 + 2x + 1) = (x - 1)(x + 1)(x + 1)^2 = (x - 1)(x + 1)^3$ .

Factor as  $(x^4 - 1) + 2x(x^2 - 1) = (x^2 + 1)(x - 1)(x + 1) + 2x(x - 1)(x + 1) = (x - 1)(x + 1)[(x^2 + 1) + 2x] = (x - 1)(x + 1)(x + 1)^2 = (x^2 - 1)(x + 1)^2$ . So roots:  $x = 1$  (from  $x^2 - 1$ ),  $x = -1$  (double, from both factors).

*Alternatively:* write  $(x^4 - 1) + 2x(x^2 - 1) = (x^2 - 1)(x^2 + 1 + 2x) = (x - 1)(x + 1)(x + 1)^2$ . Roots:  $x = 1$  and  $x = -1$  (multiplicity 3? — but we want distinct reals). Since  $(x + 1)^2(x - 1)(x + 1) = (x + 1)^3(x - 1)$ , roots are  $x = 1$  and  $x = -1$ .

*Correct factoring:*  $x^4 + 2x^3 - 2x - 1 = (x^4 - 1) + 2x(x^2 - 1) = (x^2 - 1)(x^2 + 1) + (x^2 - 1)(2x) = (x^2 - 1)(x^2 + 2x + 1) = (x - 1)(x + 1)(x + 1)^2 = (x - 1)(x + 1)^3$ . Real roots:  $x = 1, x = -1$ .

**Investigate further:** The factorisation  $x^4 + 2x^3 - 2x - 1 = (x - 1)(x + 1)^3$  shows  $x = -1$  is a root of multiplicity 3. Verify by computing  $f(-1)$ ,  $f'(-1)$ ,  $f''(-1)$  and checking they are all zero. Geometrically, a root of multiplicity 3 means the curve  $y = f(x)$  crosses the  $x$ -axis and has an inflection point there simultaneously. Explore: what does multiplicity 4 look like? Sketch  $y = (x - a)^4$  near  $x = a$ .

2. Prove that  $n^2 + n + 41$  is not always prime by finding the smallest counterexample, and explain why it is prime for  $n = 0, 1, \dots, 39$ .

**Answer:** Smallest counterexample:  $n = 40$ , giving  $40^2 + 40 + 41 = 41^2 = 1681 = 41 \times 41$ .

**Method:**  $n = 40$ :  $1600 + 40 + 41 = 1681 = 41^2$ . Composite.  $\square$  For  $0 \leq n \leq 39$ : one approach is to observe that  $f(n) = n^2 + n + 41$  — any factor  $d$  of  $f(n)$  with  $d \leq 40$  would divide  $f(n) - f(0) = n(n + 1)$  and also 41, forcing  $d = 1$  or  $d = 41$ . Since  $f(n) < 41^2$  for  $n < 40$ , the factor 41 cannot appear.

**Investigate further: Euler's lucky numbers.** The polynomial  $n^2 + n + p$  is prime for  $n = 0, 1, \dots, p - 2$  if and only if  $p$  is one of Euler's lucky numbers: 2, 3, 5, 11, 17, 41. This connects to the theory of *imaginary quadratic fields* with class number 1. Verify that  $n^2 + n + 17$  is prime for  $n = 0, \dots, 15$  and find the first counterexample. Then show directly that  $n^2 + n + 41 \equiv 0 \pmod{41}$  iff  $41 \mid n$  or  $41 \mid (n + 1)$ .

3. Let  $a, b, c > 0$  with  $abc = 1$ . Prove that  $a + b + c \geq 3$ .

**Answer:** Proof via AM-GM: see method.

**Method:** By AM-GM on three positive numbers:  $\frac{a + b + c}{3} \geq \sqrt[3]{abc} = \sqrt[3]{1} = 1$ . Therefore  $a + b + c \geq 3$ . Equality iff  $a = b = c = 1$ .  $\square$  This is the AM-GM inequality in its three-variable form, with the constraint doing all the work.

**Investigate further: AM-GM for  $n$  variables:**  $\frac{a_1 + \dots + a_n}{n} \geq \sqrt[n]{a_1 \dots a_n}$  for positive reals, with equality iff all are equal. Use it to prove: for  $a, b, c > 0$  with  $a + b + c = 3$ , we have  $abc \leq 1$ . (*This*



is the ‘flipped’ version of today’s problem.) Then prove:  $a^2 + b^2 + c^2 \geq \frac{(a+b+c)^2}{3}$  (power-mean inequality) using  $3(a^2 + b^2 + c^2) \geq (a+b+c)^2$  via the Cauchy-Schwarz inequality.

4. For positive reals  $a, b$ , prove  $\frac{a}{b} + \frac{b}{a} \geq 2$ . Hence prove AM-GM:  $\frac{a+b}{2} \geq \sqrt{ab}$ .

**Answer:** Proof: see method.

**Method:** *Part 1:* Let  $t = \frac{a}{b} > 0$ . Then  $t + \frac{1}{t} - 2 = \frac{t^2 - 2t + 1}{t} = \frac{(t-1)^2}{t} \geq 0$ . So  $\frac{a}{b} + \frac{b}{a} \geq 2$ , equality iff  $a = b$ .  $\square$

*Part 2:* Let  $a = x^2, b = y^2$  (so  $x, y > 0$ ). Part 1 gives  $\frac{x^2}{y^2} + \frac{y^2}{x^2} \geq 2$ . Instead use directly:  $(\sqrt{a} - \sqrt{b})^2 \geq 0 \Rightarrow a + b \geq 2\sqrt{ab} \Rightarrow \frac{a+b}{2} \geq \sqrt{ab}$ .  $\square$

**Investigate further:** These two proofs show that *every algebraic inequality ultimately rests on the fact that squares are non-negative*. Find at least three distinct proofs of AM-GM for two variables. Then prove the *weighted AM-GM*: for  $\lambda \in (0, 1)$ ,  $\lambda a + (1 - \lambda)b \geq a^\lambda b^{1-\lambda}$ . (This generalises AM-GM and is fundamental in convex analysis.)

5. Find the number of integers  $n$  with  $1 \leq n \leq 1000$  such that  $(n+3) \mid (n^2 - 1)$ .

**Answer:** 7

**Method:**  $n^2 - 1 = (n+3)(n-3) + 8$ . So  $(n+3) \mid (n^2 - 1)$  iff  $(n+3) \mid 8$ . The divisors of 8 are 1, 2, 4, 8. Since  $n \geq 1$ , we have  $n+3 \geq 4$ , so valid divisors of 8 that are  $\geq 4$ :  $\{4, 8\}$ .  $n+3 = 4 \Rightarrow n = 1$ .  $n+3 = 8 \Rightarrow n = 5$ .

giving  $n = -4, -5, -7, -11$  (all outside range). Also  $n+3 = 1 \Rightarrow n = -2$  (out of range);  $n+3 = 2 \Rightarrow n = -1$  (out of range). So within  $1 \leq n \leq 1000$ : only  $n = 1$  and  $n = 5$ .

*Re-examine:* divisors of 8 that are positive and  $\geq 4$ : 4 and 8. So only **2** values:  $n = 1$  and  $n = 5$ .

**Investigate further:** Generalise: find all  $n \geq 1$  such that  $(n+k) \mid (n^2 - 1)$  for fixed  $k$ . Write  $n^2 - 1 = (n+k)(n-k) + (k^2 - 1)$ , so the condition becomes  $(n+k) \mid (k^2 - 1)$ . This is a standard technique: *reduce the divisibility to a condition on a constant*. Use it to find all  $n \geq 1$  with  $(n+2) \mid (n^2 + 1)$ .

## Top 5 Patterns Today

- Remainder & Factor Theorem.** To find the remainder of  $f(x) \div (x-a)$ : evaluate  $f(a)$ . To show  $(x-a)$  is a factor: show  $f(a) = 0$ . Instantaneous — no long division.
- AM-GM and the non-negativity of squares.** Every algebraic inequality ultimately reduces to  $(x-y)^2 \geq 0$  or a sum-of-squares. AM-GM ( $\frac{a+b}{2} \geq \sqrt{ab}$ ) is the compressed form of this for products.
- Completing the square in two (or more) variables.**  $(x+h)^2 + (y+k)^2$  is the canonical form for optimisation. The minimum is the squared distance from a point to the origin after shifting.
- Rational Root Theorem.** For  $a_n x^n + \dots + a_0$ , rational roots have the form  $p/q$  with  $p \mid a_0$ ,  $q \mid a_n$ . Combined with the Factor Theorem: test candidates, divide out, repeat.
- Divisibility via algebraic identity.** To show  $(n+k) \mid f(n)$ : write  $f(n) = Q(n) \cdot (n+k) + R$

where  $R$  is a constant. Then  $(n+k) \mid f(n)$  iff  $(n+k) \mid R$  — reducing an infinite problem to a finite one.

### Common Traps to Avoid

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- 1. Applying AM–GM to negative numbers.**  $\frac{a+b}{2} \geq \sqrt{ab}$  requires  $a, b > 0$ . If  $a < 0$ , the square root  $\sqrt{ab}$  may not even be real. Always check the sign condition before applying any mean inequality.
- 2. Introducing extraneous solutions when squaring.** Squaring  $\sqrt{A} = B$  requires  $B \geq 0$  *first*. Always substitute back into the original equation to check.
- 3. Forgetting that the Rational Root Theorem gives candidates, not roots.** Every candidate must be tested. Not every  $p/q$  is a root — the theorem only rules out impossible rational roots, it does not guarantee any exist.
- 4. Confusing divisibility direction.**  $(n+k) \mid (n^2-1)$  does *not* mean  $(n^2-1) \mid (n+k)$ . Write out the algebra explicitly:  $a \mid b$  means  $b = ka$  for some integer  $k$ .
- 5. Treating multiplicity- $k$  roots as  $k$  distinct solutions.**  $x = -1$  is a root of  $(x+1)^3(x-1) = 0$  with multiplicity 3 but it is *one* value of  $x$ . Count solutions, not multiplicities, unless the question asks.

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*“Mathematics is not about numbers, equations or algorithms: it is about understanding.”*  
 — William Paul Thurston

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other prep to look at today:  
 Daily Integral for Calculus and Logic